

Computer Architecture Suggestions

CSE 2105, Date: 02.06.2024

Examples of all chapters! ***

Chapter 1:

- 1.3, Review Questions x 5
- Draw and briefly discuss the top level structure of a computer. Fig 1.4

Chapter 2:

- Review Questions: 2.1 to 2.6
- What is ARM Processor? Discuss the usage of ARM Processor. (pg 48)
- What do you know about embedded system? Discuss the marketplace where embedded system is used. With example. (pg 47)
- Different types of Microprocessors. Like 8086, Pentium.... all, see short notes
- Which types of techniques are used to improve the speed of microprocessor?
- Different generations of computer.
- Describe Von Newman Machine with proper figure.
- Describe the IAS Computer, different types of registers used in IAS Computer.
- Moore's Law
- Discuss the Design the IAS Computer's Memory Formats.

Chapter 3:

- EDVAC Computer's basic components.
- Different classes of computer. [Short Notes] - Mainframe, Minicomputer, ...
- Response Time, Throughput, Latency - definitions.
- What is IC? Why we use it?
- Which items must be focused on while designing a computer? [Instruction Set, Architecture]
- Design today for tomorrow's market. Which trends [HW+SW] should be followed to design a computer.
- Maths***

Chapter 4:

- Discuss CPU's different components.
- PC, MBR etc. definitions.
- Maths***
- Write the possible reasons why newer generation's CPU are becoming faster?
- Discuss CPU's instruction cycle. (2.3) *** [Fig not needed]
- Draw Fig 2.3 [Memory Connected CPU]

Chapter 5:

- RISC vs CISC
- Maths
- How many addressing modes are there? [3, 2, 1, 0]
- Check the maths. Convert between these.
- Register-Register, Memory-Memory, Register-Memory, Memory-Register : Advantages and Disadvantages.

Chapter 6:

- Addressing Modes, all problems.
- Draw Fig 4.1-2 [page 59]

Chapter 7:

- All example aren't required.
- Draw a general view of Finite State Machine where combination logic and hardware as attached. [Fig 5.7]
- Discuss CPU Load and Store instructions! Memory read/write. Page 90
- Example 5.2-3
- How can we evaluate/find out branch condition? How many ways? [Condition Code, Condition Register, Compare and Branch], explain these. Page-82

Chapter 8:

- Write down the situations where computer interrupts are used?
- Classifications, discuss. [6.2]
- Which types of actions are taken by control unit to manage interrupts? How to track interrupts.
- Figures are not needed.

- MIPS*** [6.6], the registers. Structural discussion. Three architectures.

Chapter 9:

- Math examples.
- Explain the Principle of locality of memory hierarchy.
- What is Memory hierarchy? Different levels of this. What's happening in each step of it?
- Hit, Miss. Rates. Definitions.
- Describe the relations between “miss rate + miss penalty” and block size. With figure. [page 130] Check online!

Chapter 10:

- Cache Memory Details.
- Write down the block placing techniques in cache. [Three Types], 8.2
- Problems of Direct Map and Fully Associative Mapping. How to solve using other techniques?
- No math!
- Describe the replacing algorithms of cache memory. [Three techniques], No Figure.
- Describe the Cache write policies.
- Cache Performance, Math [8.5-8.7]

Extra:

- Why memory hierarchy needed?
- How the hierarchy organized?
- What are the methods of accessing methods of computer memory.
- Different types of ROMs and RAMs. [DRAM vs SRAM], ROM vs RAM, ROM vs ROM, types - short notes.
- Cache memory policies, three mappings' advantages disadvantages.
- What is CPU register?
- Discuss the role of Control and Status register in CPU.
- Extra Slide: Instruction Cycle State Diagram. [Fig 3.6]
- List and briefly define the possible states that define an instruction cycle state.